
DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

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ABSTRACT

AI weather models are often reported to underestimate extremes, but most evidence concerns deterministic regression models verified against reanalysis. We evaluate twelve physical and AI forecast models against ECMWF IFS using ten months of European station observations. The evaluation covers 10 m wind, 2 m temperature, solar radiation, and precipitation, with observed-value regimes defined from a fixed ERA5 1991–2020 climatology and skill computed on matched samples. We find no uniform AI-specific deficit in the tails. Several AI models remain more accurate than IFS under extreme conditions, while others deteriorate markedly; comparable variation is present among physical models. These model-specific differences persist across variables and supporting lead-time and regional comparisons. At the same time, every model exhibits a common observation-conditioned error pattern, overpredicting low values and underpredicting high values. Attenuation of extreme observations therefore does not imply a uniform loss of relative skill: performance in the tails depends on the individual model, variable, and evaluation setting rather than on whether a forecast is produced by AI or physical numerical modelling.

1 INTRODUCTION

Data-driven weather models now outperform operational physical models on many standard forecast metrics (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). Their performance at extremes, however, remains an open concern. Models trained with mean-squared error tend towards the conditional mean, which can smooth sharp gradients and attenuate rare events (Olivetti & Messori, 2024a; Bonavita, 2024). Existing evaluations support this concern, reporting underestimated peak winds, hot and windy tails, and record-breaking events (Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

The available evidence has two important limitations. First, it is dominated by first-generation deterministic regression models. Generative ensembles, probabilistically trained models, and high-resolution regional AI models have received much less attention. Second, most studies verify forecasts against reanalysis rather than direct station observations. Whether their conclusions extend to newer model classes and independent in situ measurements is therefore unclear.

Two questions are often conflated in this discussion: whether forecasts systematically attenuate extreme observations, and whether AI models are less accurate than physical models under the same conditions. The first concerns absolute forecast error; the second concerns relative skill. A model may underestimate an extreme event while still producing a smaller error than its physical counterpart.

We examine these questions using thirteen forecast models, including ECMWF IFS as the reference. Forecasts are verified over ten months against European synoptic, solar-radiation, and rain-gauge observations. The evaluation covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly precipitation at lead times up to 48 h. Models are compared both overall and within regimes defined by the observed value relative to a fixed 1991–2020 climatology.

Throughout the paper, “extreme” refers to the tails of this local climatological distribution, principally observations below the 5th or above the 95th quantile. These regimes identify conditions that are unusual for a given location, calendar day, and UTC hour; they are not necessarily record-breaking or impact-defined events. Precipitation is treated separately within the same framework by distinguishing dry hours and quantiles of the wet-hour distribution (Zhang et al., 2011; Perkins & Alexander, 2013).

The analysis reveals a common tendency among AI and physical models to overpredict low observations and underpredict high observations. Relative skill within those regimes, however, varies substantially by model and variable. This distinction is central: attenuation of extreme values does not imply a uniform AI-specific deficit relative to physical forecasting.

The study makes three contributions:

- It extends station-based evaluation of weather extremes to deterministic regression models, a CRPS-trained global ensemble, and high-resolution regional generative models.
- It separates relative MAE within climatological regimes from the conditional forecast errors observed in those regimes.
- It tests whether these patterns persist across four surface variables, forecast lead times, countries, and model classes, with additional categorical evaluation of heavy precipitation.

Section 3 describes the verification protocol.

2 RELATED WORK

Evaluations of early AI weather models report competitive average skill alongside attenuated extremes. FourCastNet, Pangu-Weather, and GraphCast underestimated peak winds during storm Ciarrán despite reproducing its track and central pressure (Charlton-Perez et al., 2024). Broader studies similarly identify deficits in hot, windy, and record-breaking events, particularly at longer lead times (Olivetti & Messori, 2024b; Zhang et al., 2026). These errors have been linked to reduced effective resolution and the suppression of sub-synoptic variability (Bonavita, 2024). However, station-based evidence suggests a more qualified picture: performance can remain competitive for moderate extremes while degrading for the most intense events (Ben Bouallègue et al., 2024). Existing evidence is therefore concentrated on deterministic regression models evaluated primarily against reanalysis.

Recent work indicates that attenuation may depend on the training objective rather than on data-driven forecasting itself. The grid-point mean-squared-error objective can produce a double penalty that reduces effective resolution, while amplitude-preserving objectives improve surface-wind extremes (Subich et al., 2025). Diffusion-based models likewise preserve flow statistics that ensembles of regression models tend to suppress (Molinaro et al., 2024). At weather scales, GenCast improves on the ECMWF ensemble for ERA5-defined extremes, and StormCast demonstrates generative modelling at convection-allowing resolution (Price et al., 2025; Pathak et al., 2026). Direct station-based comparisons across regression, probabilistic, generative, and physical models nevertheless remain limited.

WeatherReal, MAUSAM, and StationBench provide observational benchmarks for data-driven weather forecasting (Jin et al., 2024; Gupta et al., 2025; Schlueter & Wagner, 2025). Their results show that verification against reanalysis can overstate performance relative to in situ observations. We build on this work by comparing climatological tail skill across model classes using independent European station observations.

3 METHODS

3.1 MODELS

We evaluate twelve forecast models against the ECMWF IFS high-resolution deterministic forecast, which serves as the reference (Table S1). The comparison spans physical and AI models, global and regional domains, and deterministic and ensemble forecasts.

The physical models are ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models comprise the global regression models ECMWF AIFS, Microsoft Aurora, and Jua EPT-2 Reasoning; the global ensembles ECMWF AIFS ENS and Jua EPT-2e; the regional generative ensembles Jua EPT-2 HRRR and EPT-2.1 Europa; and the solar-specialised regional model EPT-2.1 Helios (Lang et al., 2024a; Bodnar et al., 2025; Molinaro et al., 2025; Lang et al., 2024b; Siegenheim, 2025; Jua AI, 2026; Molinaro, 2026). AIFS ENS retains the AIFS architecture but is trained with a continuous ranked probability score objective and learned noise injection; it is evaluated as a 51-member ensemble. Related generative methodology is described by Molinaro et al. (2026; 2024).

Models are evaluated only for variables with compatible outputs. Precipitation is available from the physical models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS ENS is restricted to 10 m wind and 2 m temperature because its radiation and precipitation outputs cannot be matched to the hourly accumulations used for verification.

3.2 STATION OBSERVATIONS AND EVALUATION

Station data. Forecasts are verified against three complementary in situ networks:

- **Synoptic:** quality-controlled observations of 10 m wind and 2 m temperature.
- **Solar radiation:** hourly global horizontal irradiance from DWD, Météo-France, MeteoSwiss, KNMI, RMI, DMI, MET Norway (Frost), GeoSphere Austria, SMHI, FMI, BSRN, and ARPA Lombardia.
- **Precipitation:** hourly accumulations from 4,005 rain gauges operated by DWD, Météo-France, MET Norway, AEMET, GeoSphere Austria, the UK Met Office, KNMI, RMI, CHMI, DMI, and MeteoSwiss. This separate network is required because the synoptic feed does not provide usable hourly precipitation accumulations.

The evaluation covers thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. The precipitation network excludes Italy and Poland. Forecasts are bilinearly interpolated to observation locations (Molinaro et al., 2026). Precipitation is scored only at gauges within each model’s domain, and regional comparisons use a common gauge sample within each evaluation cell.

None of the Jua models was trained or fine-tuned on these observations. Historical station errors enter the evaluation only through the causal bias correction described in Section 3.4.

Evaluation tracks. Two tracks account for differences in model availability and output cadence.

- **Track A** covers 1 September 2025 to 30 June 2026. Wind and temperature are evaluated for IFS and the ten comparator models available throughout this period; ICON-EU is reserved for Track B. Leads range from 6 to 48 h at 6-hourly intervals, matching the cadence of AIFS, AIFS ENS, and Aurora.
Solar radiation is evaluated over the same period at hourly leads from 1 to 48 h. EPT-2.1 Helios is included, while AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. The solar network contains no stations in GB, ES, or PL. Italian coverage, provided mainly by ARPA Lombardia, ends in February 2026; subsequent solar results therefore exclude Italy.
Precipitation is evaluated from 1 to 48 h for models with compatible output. Hourly models contribute at every lead, whereas ECMWF ENS contributes only at its native 6-hourly cadence.
- **Track B** covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU over a common hourly lead grid from 1 to 48 h. All four variables are evaluated, with EPT-2.1 Helios additionally included for solar radiation.

3.3 OBSERVED-VALUE REGIMES

Regimes are defined from the fixed 1991–2020 ERA5 climatology (Hersbach et al., 2020), independently of the evaluation period. ERA5 fields are bilinearly interpolated from the 0.25° grid to each

station. For each station, calendar day, and UTC hour, the reference distribution comprises values from a cyclic ± 15 -day window over the 30-year period. This yields 930 values before variable-specific conditioning and preserves seasonal and diurnal variability.

Wind and temperature use the complete reference distribution. Solar radiation and precipitation require additional conditioning because both contain a structural mass at zero. Solar thresholds are estimated from positive daytime GHI. For precipitation, observations below 0.1 mm form a separate dry regime, and the remaining thresholds are estimated from wet hours.

Let $q_\alpha(s, d, h)$ denote the α -quantile of the relevant reference distribution at station s , calendar day d , and UTC hour h . Wind, temperature, and solar radiation use $q_{0.05}$, $q_{0.25}$, $q_{0.75}$, and $q_{0.95}$. Precipitation uses $q_{0.50}^{\text{wet}}$, $q_{0.75}^{\text{wet}}$, and $q_{0.95}^{\text{wet}}$. We omit the dependence on station, day, and hour below.

Each forecast–observation pair is assigned to a regime according to its observed value y , so membership is identical across models and lead times. Tables S2 and S3 give the corresponding intervals and meteorological labels.

The use of a fixed local climatology follows established practice in the climate-extremes literature (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander, 2013; Perkins, 2015); the 1991–2020 period follows the WMO climate-normal convention. Conditioning precipitation quantiles on wet hours is consistent with ETCCDI wet-day indices such as R95p.

Because thresholds are estimated independently of the evaluation period, observed regime frequencies need not equal their nominal probabilities. Moreover, ERA5 grid-cell values have lower spatial variance than point observations, producing broader wind and solar tails than a station-based climatology. The primary regimes of interest are the upper wind and temperature tails, both solar tails, and the upper wet-hour precipitation tail.

3.4 BIAS CORRECTION

We apply a causal rolling bias correction before computing the reported scores. For wind, temperature, and daytime solar radiation, the correction is additive. Let w denote a valid week, h the UTC hour, and $\mathcal{D}_{w,h}$ the matched forecast–observation pairs at hour h during the preceding four weeks. For model m ,

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{w,h}|} \sum_{(f_m, y) \in \mathcal{D}_{w,h}} (f_m - y), \quad \tilde{f}_m(t) = f_m(t) - \bar{b}_{m,w,h(t)}. \quad (1)$$

The estimate is pooled across European stations and does not depend on initialisation or lead time. It is estimated separately for each model, and the same shift is applied to every ensemble member, preserving spread.

An additive correction is unsuitable for precipitation because the distribution is zero-bounded and strongly right-skewed. Let $\mathcal{P}_w$ contain precipitation forecasts and observations from the preceding four weeks, pooled across initialisation hours and lead times. Forecasts valid during week w are corrected by

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left(\frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t). \quad (2)$$

This multiplicative correction adjusts systematic wet or dry amount biases while preserving zero forecasts.

Both corrections use only information available before the valid week. Because regimes are assigned from observations, the correction does not alter regime membership. It removes recent mean bias but does not force the conditional bias within each regime to vanish.

3.5 SKILL, LEAD TIMES, AND AGGREGATION

Let m denote a forecast model and c an evaluation cell defined by track, country, variable, lead time, and regime. MAE skill relative to ECMWF IFS is

$$S_{m,c} = 1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}}. \quad (3)$$

Positive values indicate lower MAE than IFS. The numerator and denominator are computed from matched forecast–observation pairs, using a common intersection of available initialisations and the four daily cycles at 00, 06, 12, and 18 UTC.

Ensemble models are evaluated through their ensemble means. The analysis therefore compares point-forecast accuracy and does not assess ensemble spread or probabilistic calibration. Mean forecast-minus-observation error is also reported within each regime.

Results are shown at 1, 6, 12, 24, and 48 h where available and are also aggregated over the lead grid of each comparison. Track A wind and temperature use 6-hourly leads from 6 to 48 h. Track A solar radiation, Track B, and hourly precipitation models use hourly leads from 1 to 48 h. Lower-cadence models contribute only at their native leads; no temporal interpolation is applied.

Uncertainty is estimated by a leave-one-month-out jackknife and reported as one standard error. For heavy precipitation, defined by exceedance of $q_{0.95}^{\text{wet}}$, we additionally report probability of detection, false-alarm ratio, critical success index, and frequency bias. These statistics are computed from forecast and observed exceedances pooled across gauges and lead times up to 48 h.

4 RESULTS

4.1 SKILL ACROSS THE OBSERVED DISTRIBUTION

Figure 1 shows that relative skill in the tails does not follow a uniform AI–physical ordering. EPT-2.1 Europa leads the wind comparison overall and remains competitive with ICON Global at gale force. EPT-2 HRRR and EPT-2.1 Europa perform most strongly for temperature, including in the heat regime. By contrast, AIFS and AIFS ENS remain below IFS in the wind and temperature tails, despite the ensemble’s improved performance near the centre of the distribution. The largest heat deficit belongs to the physical GFS model.

Solar radiation exhibits a different pattern. EPT-2.1 Helios leads overall and under overcast and near-clear-sky conditions, but falls below IFS in the typical irradiance regime, where EPT-2 Reasoning performs most strongly. Its advantage is therefore concentrated in the tails rather than distributed uniformly across conditions.

For precipitation, the clearest gains occur in the middle of the wet-hour distribution (Figure 2). EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve substantially on IFS from moderate through high rain, whereas the physical global models remain below the reference. These gains narrow at the heaviest intensities: EPT-2 Reasoning remains slightly ahead, EPT-2.1 Europa is close to IFS, and EPT-2 HRRR and EPT-2e fall slightly below it. Europa follows a distinct pattern, deriving much of its aggregate skill from dry hours while losing skill once precipitation occurs.

4.2 SHARED CONDITIONAL-ERROR PATTERN

Despite their differences in relative skill, all models exhibit the same conditional-error structure (Figure 3): low observations are overpredicted and high observations underpredicted. The variation across regimes is considerably larger than the spread between models and is present for all four variables. A shared tendency towards the centre of the observed distribution therefore coexists with model-specific differences in relative accuracy.

Because the regimes are conditioned on the verifying observation, this pattern is not in itself evidence of model smoothing. Limited predictability, unresolved local variability, and observational or representativeness errors can produce the same structure.

4.3 ROBUSTNESS ANALYSES

The main conclusions persist across lead times (Figures S1–S4), the 1–12 h range (Figures S5 and S6), individual countries (Figures S13–S16), and the matched regional comparison (Figures S8–S12). ICON-EU is competitive with the leading AI models in the wind and temperature tails, while the principal wind, solar, and precipitation patterns remain unchanged. Complete robustness and categorical-precipitation results (Figure S7, Table S24) are reported in the supplement.

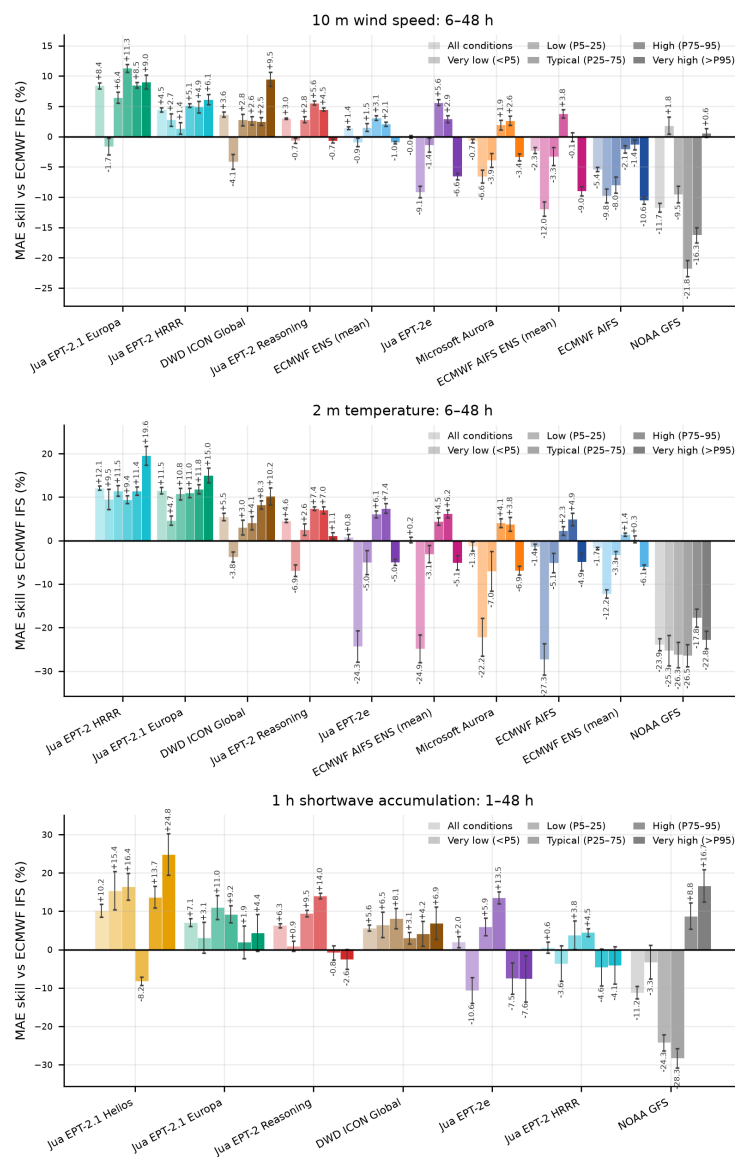


Figure 1: Track A MAE skill relative to ECMWF IFS by observed-value regime, with ± 1 jackknife SE. Wind and temperature use 6–48 h leads; solar radiation uses 1–48 h.

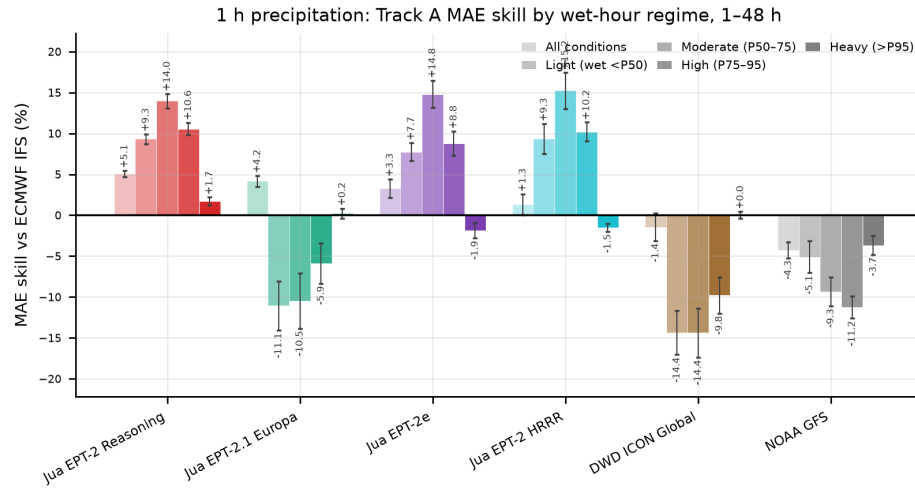


Figure 2: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime over 1–48 h, with ± 1 jackknife SE.

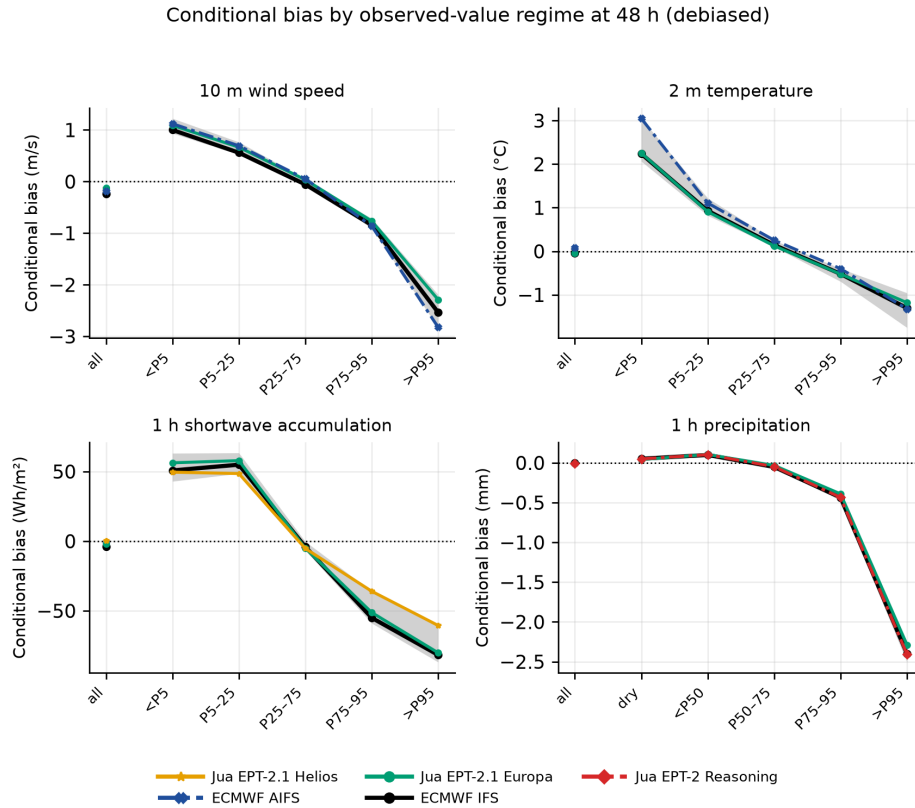


Figure 3: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models. ECMWF IFS, EPT-2.1 Europa, and ECMWF AIFS are highlighted in every panel; the solar panel additionally highlights EPT-2.1 Helios and the precipitation panel EPT-2 Reasoning.

5 DISCUSSION

The results distinguish two meanings of “missing extremes.” A forecast may underestimate the magnitude of an extreme observation, or it may be less accurate than a competing model under the same conditions. The first behaviour is common across the evaluated models: low observations tend to be overpredicted and high observations underpredicted. The second is not. Several AI models achieve lower MAE than IFS within the heat, gale, solar, and precipitation regimes. Attenuation of extreme values therefore does not imply a general AI-specific deficit in relative skill.

This distinction qualifies earlier findings based primarily on first-generation regression models verified against reanalysis (Olivetti & Messori, 2024b). The broader, station-based comparison considered here reveals substantial variation across models and variables. EPT-2.1 Helios performs strongly in both solar tails but loses skill under typical irradiance, while the Jua precipitation models improve most clearly through the middle of the wet distribution rather than at its extreme. Performance in the tails is therefore not determined by aggregate skill alone.

The results are also inconsistent with a simple AI–physical divide. The wind- and temperature-tail deficits of AIFS persist for AIFS ENS despite its improved central-regime performance. In contrast, EPT-2 HRRR and EPT-2.1 Europa remain competitive at gale force and in heat. Among the physical models, ECMWF ENS loses skill in the heat regime, GFS has the largest heat deficit, and ICON-EU is competitive with the leading regional AI models. These differences are better understood as properties of individual models than of their broad forecasting paradigm.

The comparison does not establish what causes these differences. The models vary simultaneously in architecture, training objective, resolution, training data, initialisation, forecast cadence, and spatial domain (Bonavita, 2024; Subich et al., 2025). Performance cannot therefore be attributed independently to AI versus physics, regression versus generative training, or global versus regional resolution. Moreover, evaluating ensembles through their means assesses point-forecast accuracy but not calibration, spread, or extreme-event probabilities.

Absolute attenuation and relative skill are thus complementary properties. AI models may underestimate extreme observations without being uniformly less accurate than physical models, just as strong aggregate performance does not guarantee an advantage in every tail. Within this evaluation, performance at extremes is model- and variable-specific rather than a general property of AI weather forecasting.

ETHICS STATEMENT

Several of the forecast systems evaluated in this study are developed by the authors’ organisation. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the archived requests, aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates. The package is provided as anonymised supplementary material.

AI USE STATEMENT

Generative AI tools were used to assist with software changes in the extraction, aggregation and figure code, to inspect generated results, and to draft and edit manuscript text. They were not used to generate or modify observations or forecast data. The authors checked AI-assisted code through the pipeline’s executable self-tests and regenerated artifacts, verified the quoted numbers against the aggregate data, reviewed all manuscript text, and take full responsibility for the content.

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A SUPPLEMENTARY MATERIAL

Table S1: Forecast models included in the evaluation. ECMWF IFS is the reference model.

Model	Model class	Forecast type	Domain
ECMWF IFS (reference)	Physical	Deterministic	Global
ECMWF ENS	Physical	Ensemble	Global
NOAA GFS	Physical	Deterministic	Global
DWD ICON Global	Physical	Deterministic	Global
DWD ICON-EU	Physical	Deterministic	Regional
ECMWF AIFS	Regression AI	Deterministic	Global
Microsoft Aurora	Regression AI	Deterministic	Global
Jua EPT-2 Reasoning	Regression AI	Deterministic	Global
Jua EPT-2e	Regression AI	Ensemble	Global
Jua EPT-2.1 Helios	Regression AI (solar)	Deterministic	Regional
ECMWF AIFS ENS	Probabilistic AI	Ensemble	Global
Jua EPT-2 HRRR	Generative AI	Ensemble	Regional
Jua EPT-2.1 Europa	Generative AI	Ensemble	Regional

Table S2: Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the quantiles refer to positive daytime GHI.

Regime	Definition	Wind	Temperature	Solar
All	—	all conditions	all conditions	all daytime
Very low	$y < q_{0.05}$	calm	severe cold	heavily overcast
Low	$q_{0.05} \leq y < q_{0.25}$	light air	cold	overcast
Typical	$q_{0.25} \leq y < q_{0.75}$	moderate	moderate	mixed sky
High	$q_{0.75} \leq y < q_{0.95}$	fresh to strong	warm	mostly sunny
Very high	$y \geq q_{0.95}$	gale force	heat	near clear sky

Table S3: Observed-value regimes for hourly precipitation. Wet-hour quantiles are conditional on ERA5 accumulations of at least 0.1 mm.

Regime	Definition	Interpretation
All	—	all hours
Dry	$y < 0.1 \text{ mm}$	dry hour
Light	$0.1 \text{ mm} \leq y < q_{0.50}^{\text{wet}}$	light rain
Moderate	$q_{0.50}^{\text{wet}} \leq y < q_{0.75}^{\text{wet}}$	moderate rain
High	$q_{0.75}^{\text{wet}} \leq y < q_{0.95}^{\text{wet}}$	heavy rain
Heavy	$y \geq q_{0.95}^{\text{wet}}$	most intense hours

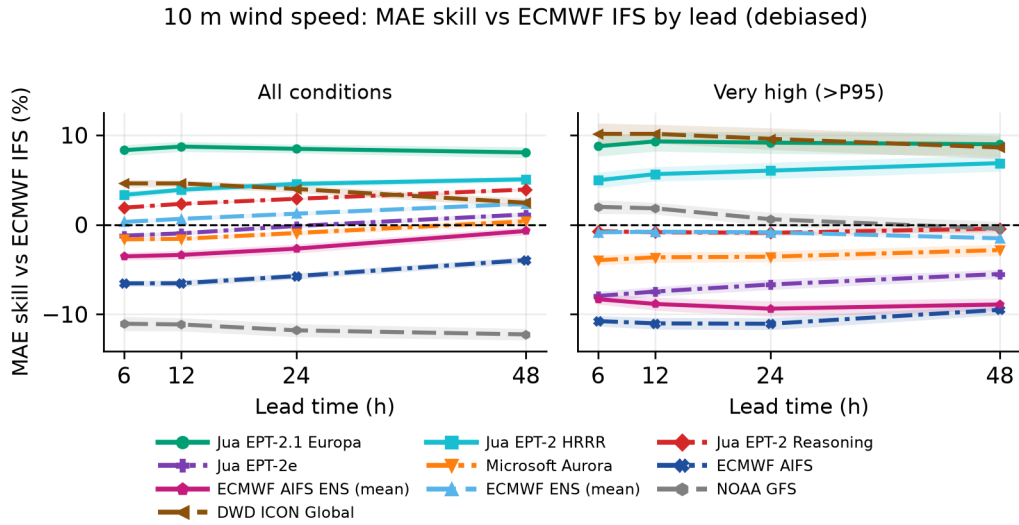


Figure S1: MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime, $y \geq q_{0.95}$. Shading denotes ± 1 jackknife SE.

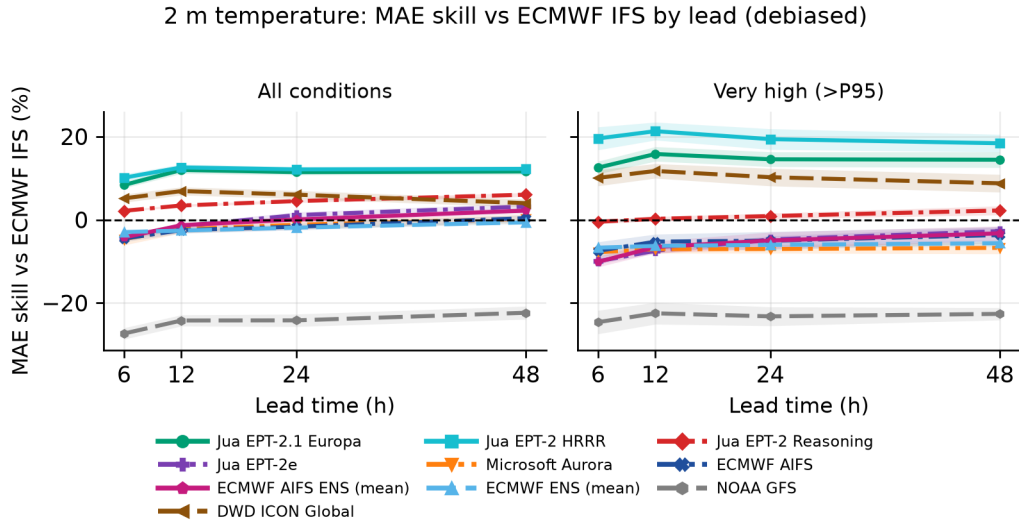


Figure S2: As Figure S1, for 2 m temperature. The right panel shows the heat regime, $y \geq q_{0.95}$.

Table S4: Track A MAE skill (%) by country in the heat regime ($y \geq q_{0.95}$), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF AIFS ENS (mean)	+5.3	-3.1	-6.7	-10.6	-12.6	-3.1	-3.9	+5.1	+4.6	-3.2	-38.7	+2.1	+4.4
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU [†]	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

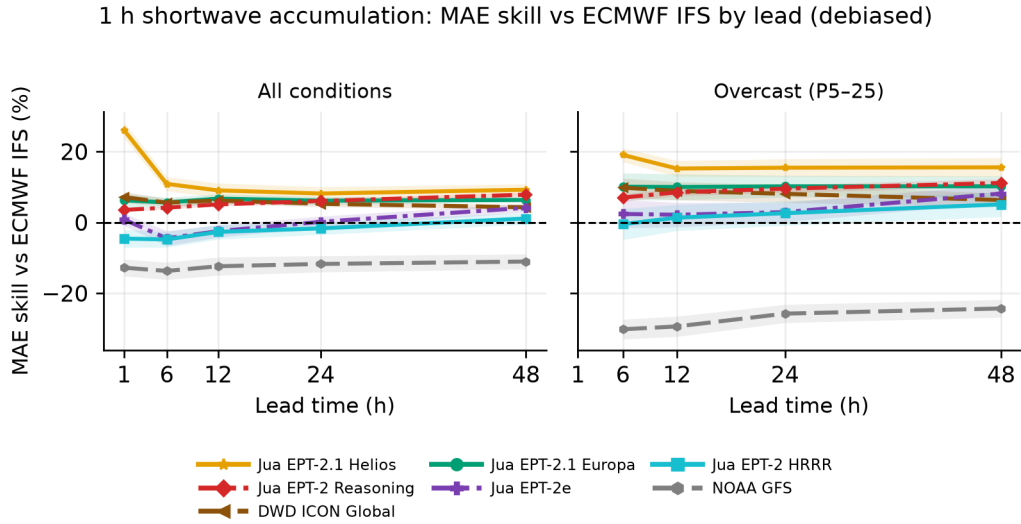


Figure S3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime, $q_{0.05} \leq y < q_{0.25}$. AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes ± 1 jackknife SE.

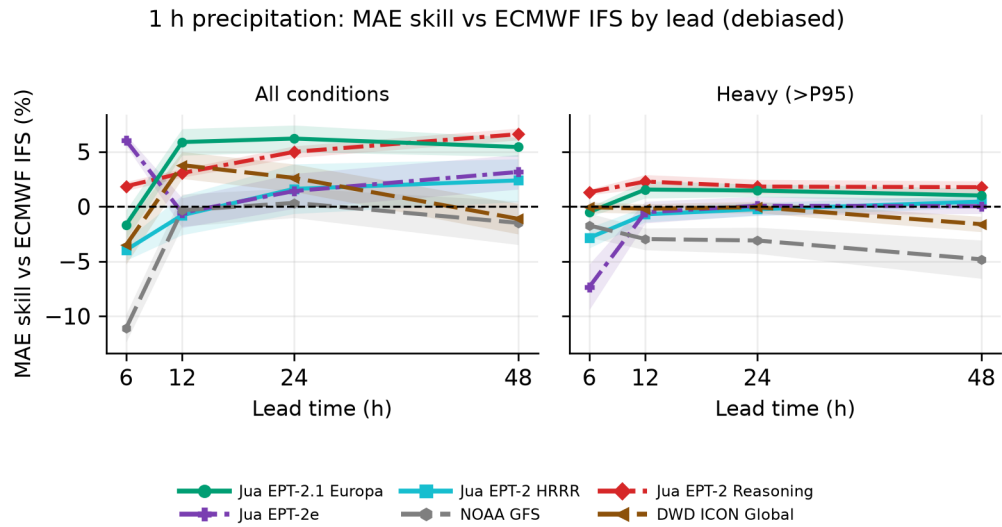


Figure S4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq q_{0.95}^{\text{wet}}$. Shading denotes ± 1 jackknife SE.

Table S5: Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (± 0.4)	+2.2 (± 1.3)	+9.3 (± 0.6)	+14.0 (± 0.9)	+10.6 (± 0.8)	+1.7 (± 0.5)
Jua EPT-2 HRRR	+1.3 (± 1.3)	-6.1 (± 3.3)	+9.3 (± 1.8)	+15.2 (± 2.2)	+10.2 (± 1.2)	-1.5 (± 0.5)
Jua EPT-2e	+3.3 (± 1.1)	+1.5 (± 4.8)	+7.7 (± 1.1)	+14.8 (± 1.7)	+8.8 (± 1.5)	-1.9 (± 0.9)
Jua EPT-2.1 Europa	+4.2 (± 0.7)	+21.0 (± 2.7)	-11.1 (± 3.0)	-10.5 (± 3.4)	-5.9 (± 2.5)	+0.2 (± 0.6)
NOAA GFS	-4.3 (± 1.0)	-0.2 (± 1.8)	-5.1 (± 2.0)	-9.3 (± 1.8)	-11.2 (± 1.4)	-3.7 (± 1.2)
DWD ICON Global	-1.4 (± 1.7)	+8.3 (± 2.7)	-14.4 (± 2.7)	-14.4 (± 3.0)	-9.8 (± 2.2)	+0.0 (± 0.4)

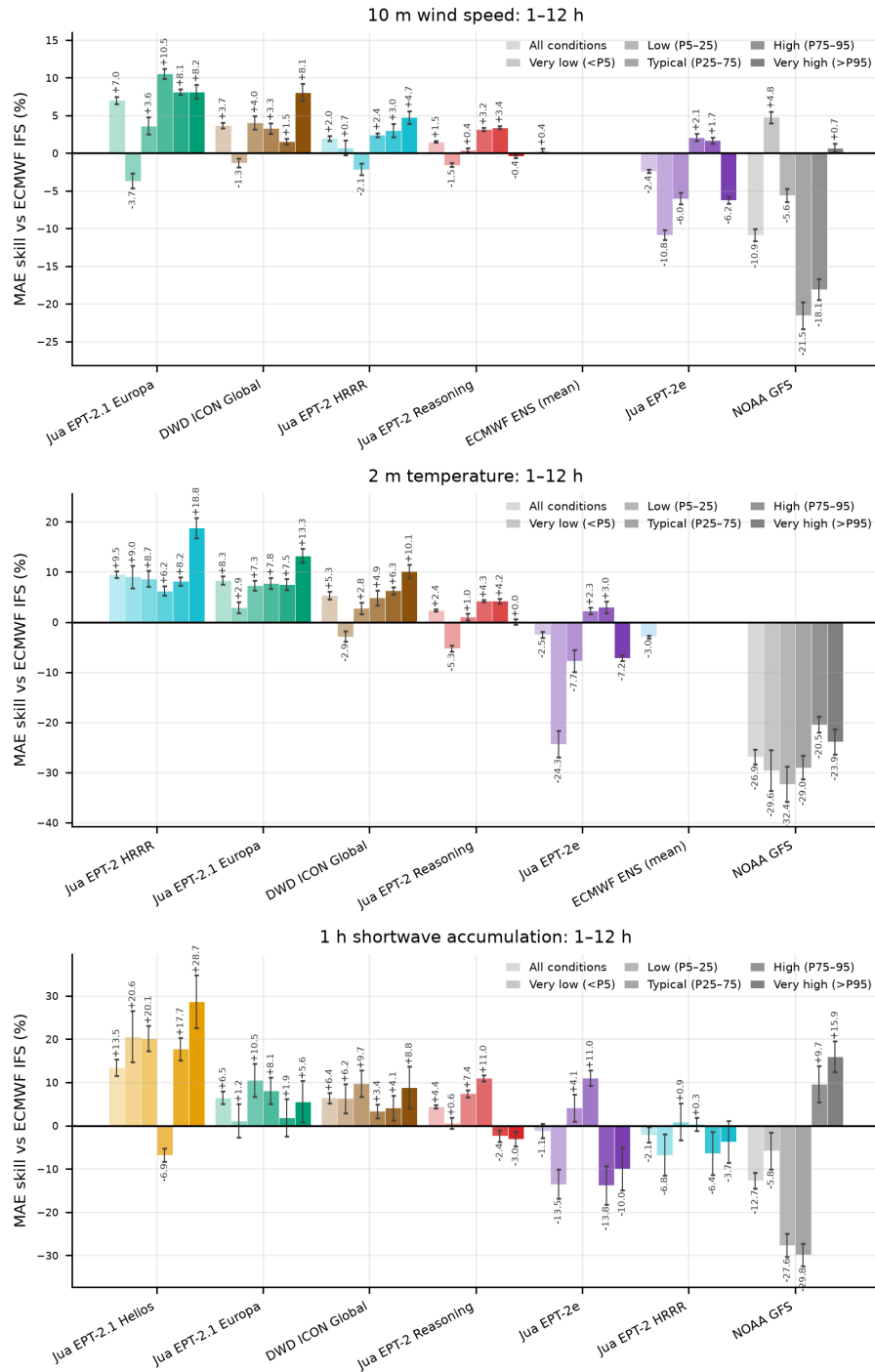


Figure S5: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with ± 1 jack-knife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

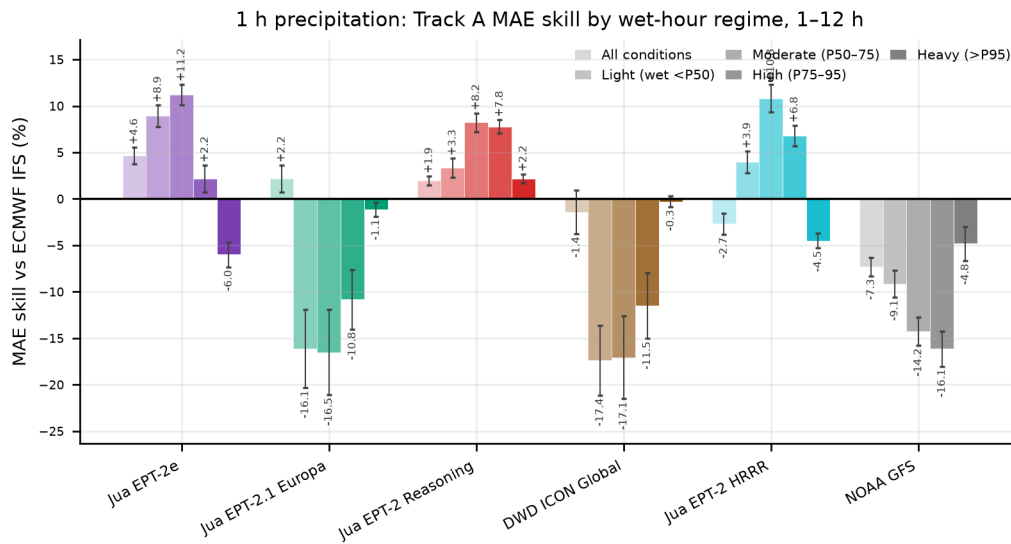


Figure S6: As Figure 2, restricted to the 1–12 h forecast range.

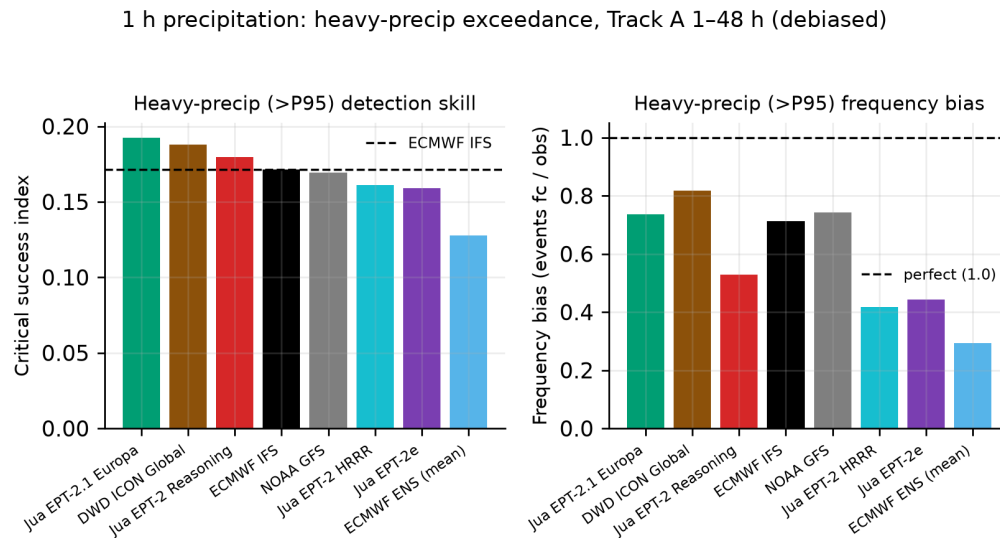


Figure S7: Detection of heavy precipitation ($y \geq q_{0.95}^{\text{wet}}$) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

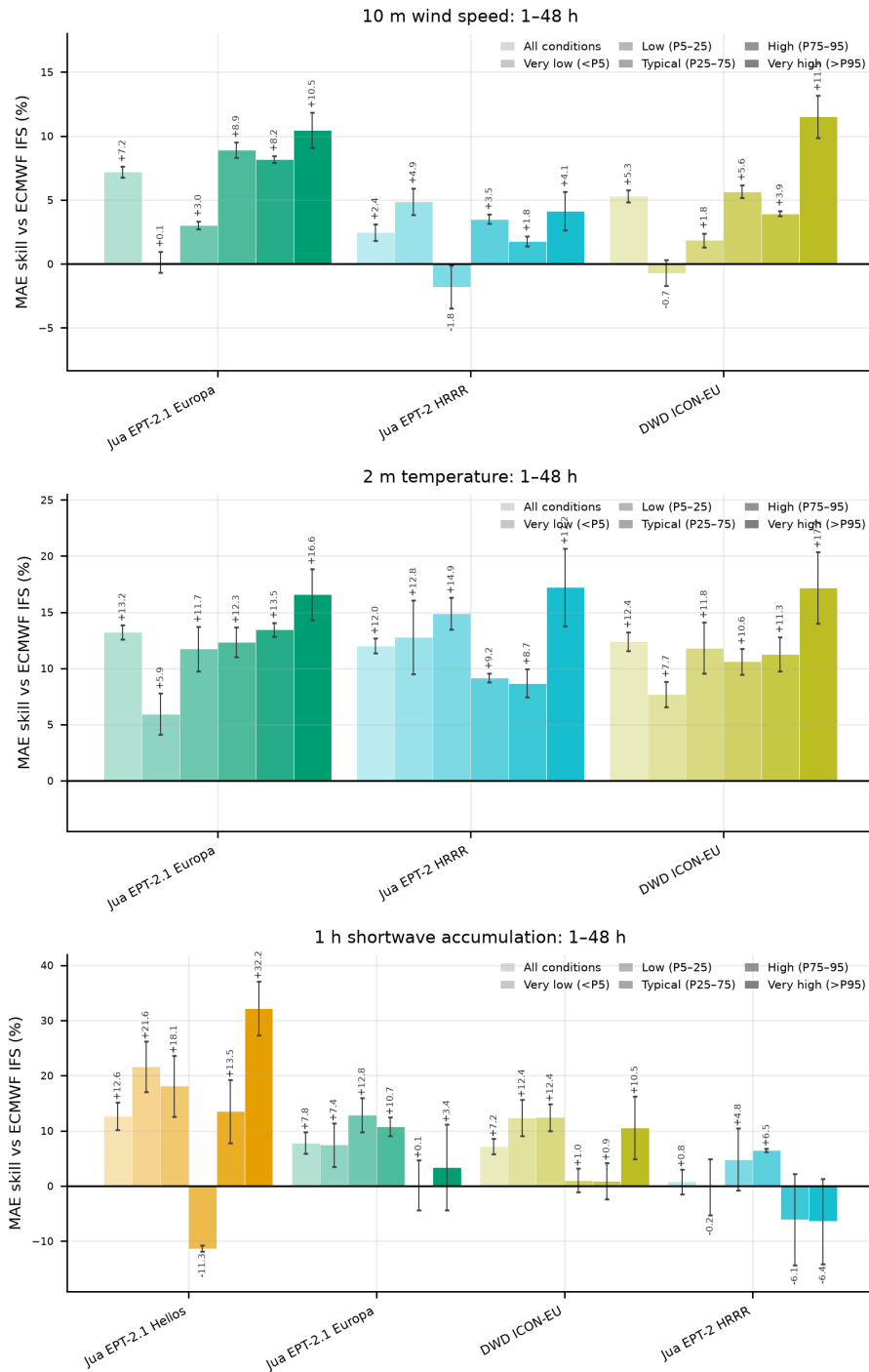


Figure S8: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

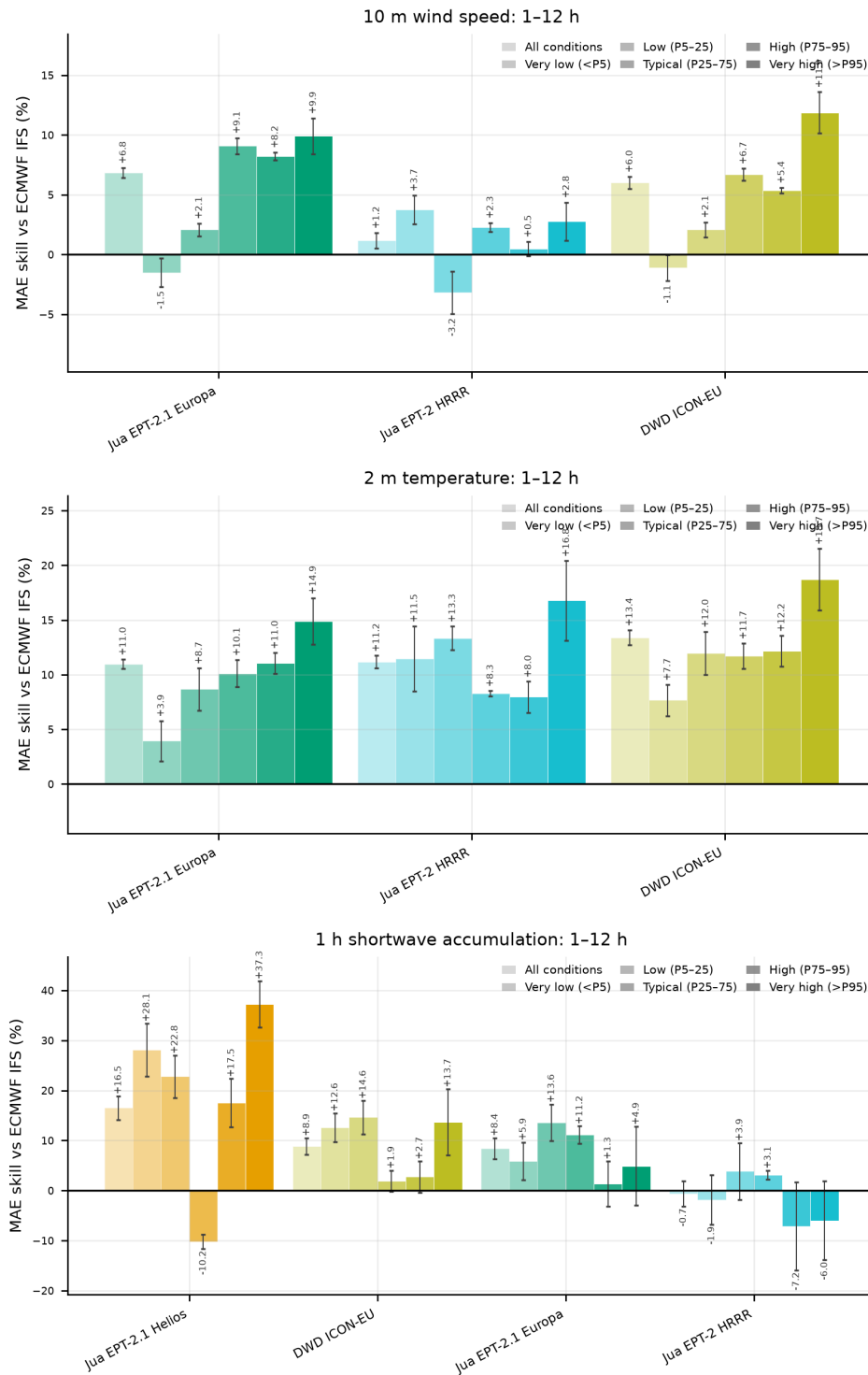


Figure S9: As Figure S8, restricted to the 1–12 h forecast range.

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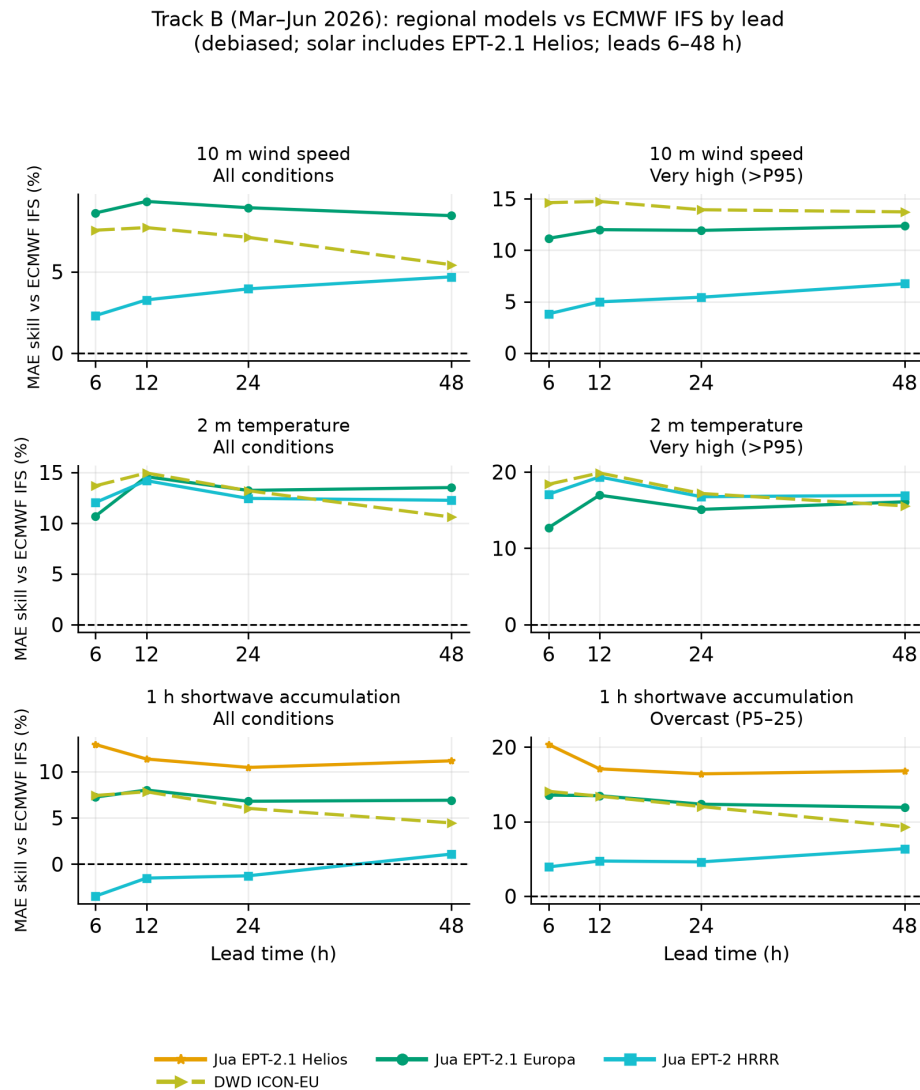


Figure S10: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes ± 1 jackknife SE.

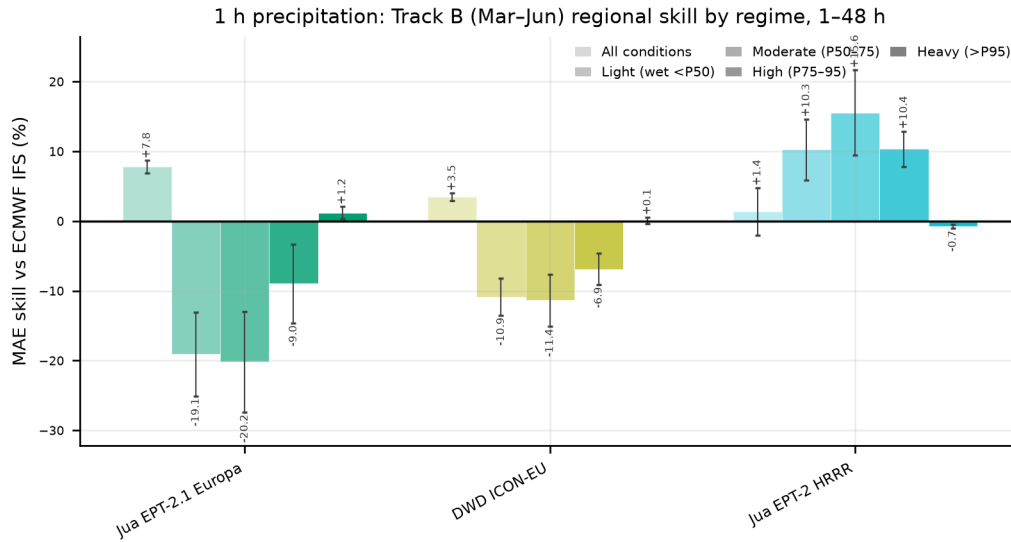


Figure S11: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE.

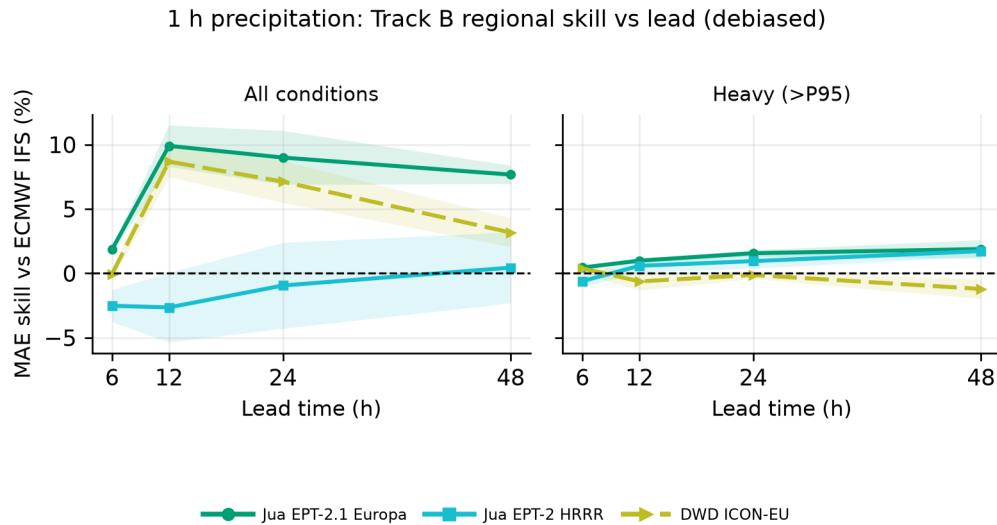


Figure S12: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq q_{0.95}^{\text{wet}}$. Shading denotes ± 1 jackknife SE.

Table S6: Track A MAE skill (%), 6–48 h, by regime – Figure 1, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF AIFS ENS (mean)	-2.3 (0.5)	-12.0 (1.2)	-3.3 (1.5)	+3.8 (0.7)	-0.1 (0.7)	-9.0 (0.7)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S7: Track A MAE skill (%), 6–48 h – Figure 1, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF AIFS ENS (mean)	+0.2 (0.6)	-24.9 (3.2)	-3.1 (2.0)	+4.5 (0.9)	+6.2 (0.9)	-5.1 (1.7)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S8: Track A MAE skill (%), 1–48 h hourly – Figure 1, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S9: Track A MAE skill (%), 1–12 h – Figure S5, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S10: Track A MAE skill (%), 1–12 h – Figure S5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S11: Track A MAE skill (%), 1–12 h – Figure S5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S12: Track A precipitation MAE skill (%), 1–12 h (Figure S6). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S13: Track A MAE skill (%) by lead – Figure S1, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
AIFS ENS	-3.5	-8.3	-3.4	-8.9	-2.7	-9.4	-0.7	-8.9
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S14: Track A MAE skill (%) by lead – Figure S2, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
AIFS ENS	-4.3	-10.1	-1.3	-6.6	+0.1	-5.0	+2.2	-3.3
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S15: Track A MAE skill (%) by lead – Figure S3, solar (Ovc. = overcast, P5–25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S16: Track B wind MAE skill (%), 1–48 h, by regime (Figure S8).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S17: Track B MAE skill (%), 1–48 h – Figure S8, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S18: Track B MAE skill (%), 1–48 h – Figure S8, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S19: Track B MAE skill (%) by lead – Figure S10, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S20: Track B MAE skill (%) by lead – Figure S10, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S21: Track B MAE skill (%) by lead – Figure S10, solar (Ovc. = overcast, P5–25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S22: Track B precipitation MAE skill (%) by lead (Figure S12). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

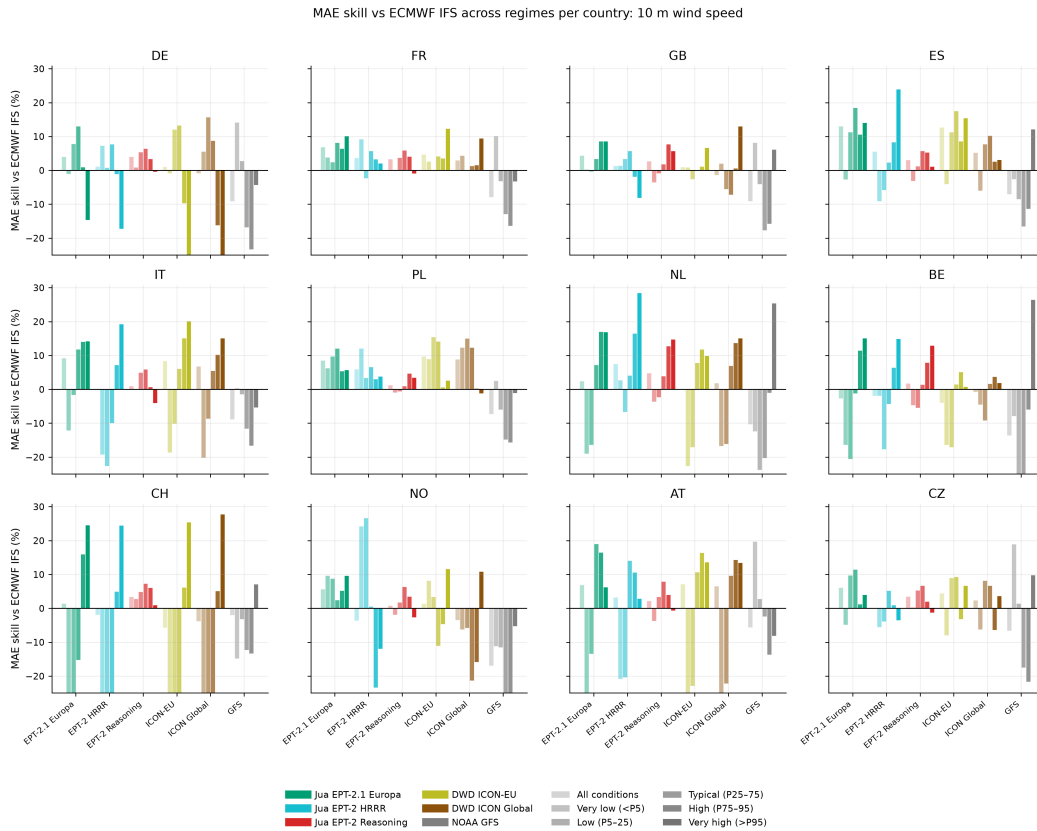


Figure S13: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set (Section 4.3).

Table S23: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure S11. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (±0.9)	+29.3 (±1.4)	-19.1 (±6.0)	-20.2 (±7.2)	-9.0 (±5.7)	+1.2 (±0.8)
Jua EPT-2 HRRR	+1.4 (±3.4)	-3.6 (±9.7)	+10.3 (±4.4)	+15.6 (±6.1)	+10.4 (±2.5)	-0.7 (±0.3)
DWD ICON-EU	+3.5 (±0.5)	+15.3 (±0.3)	-10.9 (±2.7)	-11.4 (±3.8)	-6.9 (±2.2)	+0.1 (±0.5)

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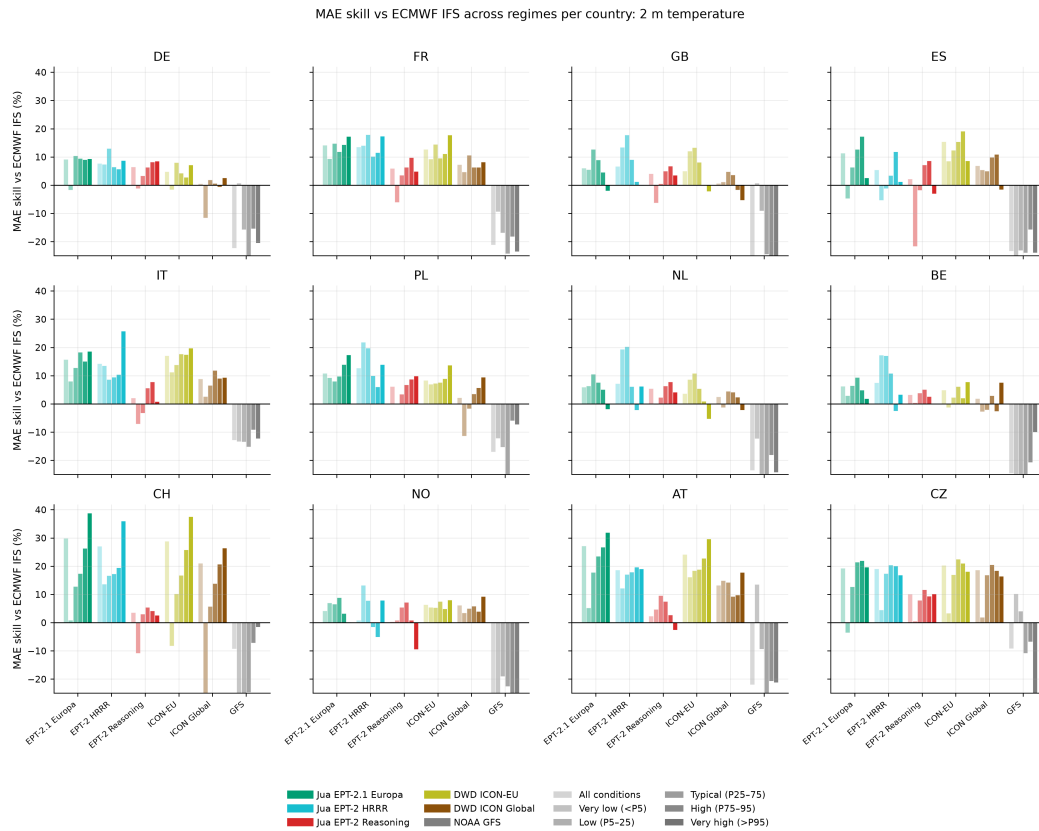


Figure S14: As Figure S13, for 2 m temperature.

Table S24: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure S7. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

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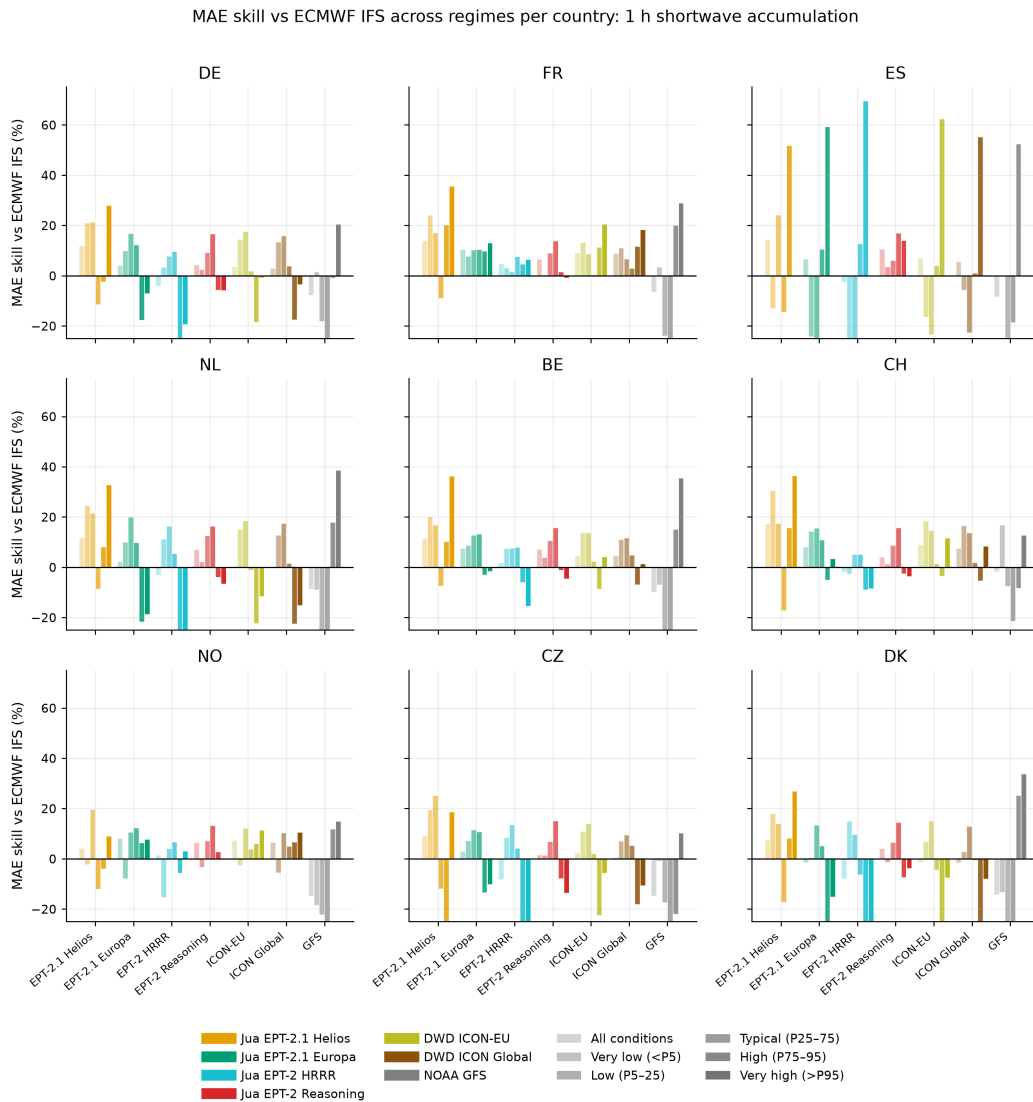


Figure S15: As Figure S13, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 3.2).

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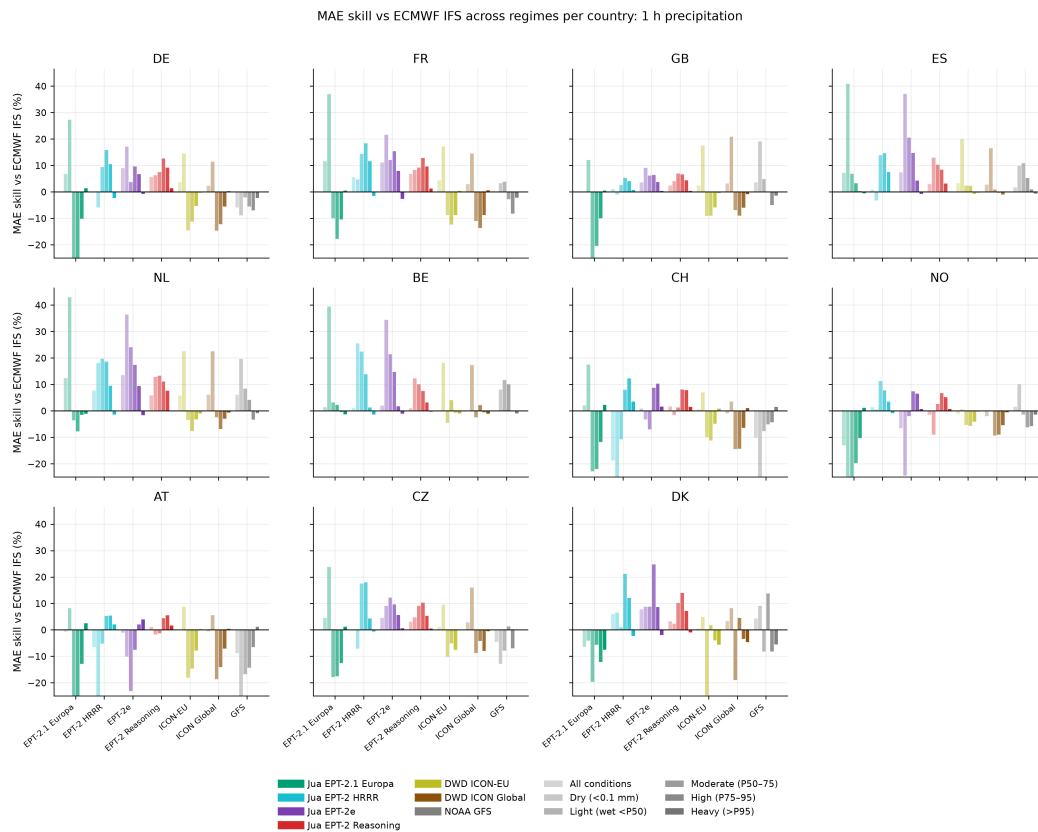


Figure S16: As Figure S13, for 1 h precipitation using the wet-hour regimes of Table S3. IT and PL have no gauges in the precipitation panel.